FinGuard AI: Localhost Development Guide

Project: Intelligent Banking Fraud Detection and Smart Transaction Protection System
Active Localhost URL: http://localhost:8501 | Final-Year B.Tech CSE (AI & ML)

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1. What Is Running on Your Localhost Right Now?

When you open **http://localhost:8501** in your web browser, you are running a complete, locally hosted AI-powered banking security system. Unlike simple UI mockups, this application is backed by an **actual trained machine learning pipeline**, a real SQLite database, and a live Dynamic Risk Engine.

Component	Current Status	What It Does
Web App (Frontend)	**Active (Port 8501)**	Streamlit web interface with interactive forms, Plotly gauges, and charts.
ML Model (Backend)	**Trained & Loaded**	Random Forest (95.4% accuracy) trained on the real IEEE-CIS Fraud dataset.
Database	**finguard.db (SQLite)**	Saves every evaluated transaction, user profile, and audit trail locally.

2. How FinGuard AI Evaluates a Transaction (5 Easy Steps)

- **Step 1: Input Received** — The system takes the transaction amount, customer profile, device status, IP network, and merchant category.
- **Step 2: ML Fraud Probability** — The trained IEEE-CIS model checks mathematical fraud signatures and outputs a probability score (0 to 100).
- **Step 3: Behavioral Analysis** — The engine checks if the amount is normal for this user (Z-Score), whether transaction frequency is too fast (1-hour velocity), and any failed PIN attempts.
- **Step 4: Contextual Gateway Risk** — It checks environmental flags: Is the user on Tor/VPN? Is the device unrecognized? Is the merchant high-risk (Crypto/Gambling)?
- **Step 5: Dynamic Risk Score & Decision** — All signals combine into a single 0–100 score: **ALLOW** (<30), **VERIFY** (30–69), or **BLOCK** (≥70).

3. The 6 Interactive Pages You Can Click On

Page Name	What to Show During Project Review / Viva
1. Overview (app.py)	Shows live system status, active architecture flow diagram, and system metrics.
2. Dashboard	Displays total transactions evaluated, fraud rate, risk distribution histogram, and decision donut chart.
3. Transaction Analyzer	**The star demo page!** Click the 6 preset buttons (Normal Coffee, Cold-Start User, Rapid Draining, Child Gaming) to see real-time score gauges and explanations.
4. Fraud Monitoring	A complete searchable audit ledger of all analyzed transactions with CSV export and forensic drill-down inspector.
5. Model Performance	Genuine academic metrics comparing Random Forest vs Gradient Boosting vs XGBoost, plus confusion matrix and top features.
6. Federated Learning	Shows simulated privacy-preserving collaboration between 3 banks (FedAvg algorithm) with zero customer data leakage.

4. How to Start and Stop the Application Anytime

- **To Open the App Right Now:** Just open your browser and go to: **http://localhost:8501**

- **To Start the App Tomorrow (One-Click):** Double-click the file named **run_app.bat** inside your project folder.
- **To Start via Terminal:** Open terminal in the project folder and type: streamlit run app.py
- **To Run the 11 Unit Tests:** Double-click **run_tests.bat** (all 11 tests pass with 100% OK).

5. Three Standout Features to Mention in Your Presentation

- **1. Cold-Start Account Handling:** When a new user has 0 historical transactions, FinGuard does not blindly block them or assign zero risk. It uses an adaptive contextual prior.
- **2. Child-Safe Spending Protection:** Detects accidental gaming/in-app purchases on parents' devices and requires parental biometric confirmation.
- **3. Explainable AI (XAI):** Every transaction gives clear human-readable bullet points explaining exactly why a transaction was allowed, challenged, or blocked.